

MARIMO AND ALTAIR

MARIMO DON'TS

- Do not re-define an object
 - Variable names and function names must not be re-used
 - Or, will cause an error like this:

This cell wasn't run because it has errors

This cell redefines variables from other cells.

SP500_data was also defined by:

cell-75

Why can't I redefine variables?

marimo requires that each variable is defined in just one cell. This constraint enables reactive and reproducible execution, arbitrary cell reordering, seamless UI elements, execution as a script, and more.

Try merging this cell with the mentioned cells or wrapping it in a function.

Alternatively, rename variables to make them private to this cell by prefixing them

Learn more at our docs ↗: <https://docs.marimo.io/guides/reactivity/#multiple-definition-error>

ALTAIR VS. MATPLOTLIB

- `Matplotlib` (not covered in this module) is used as a benchmark to contrast with `Altair`

MATPLOTLIB

- Is both the oldest and the most widely used visualization library
- An **imperative** library: Focus on explicit, step-by-step commands to achieve a goal
- Offer **granular control** over everything. For example,
 - Can explicitly command the library to “change this specific tick’s color”, “move this label 2 pixels left”, etc
- Limitations include:
 - **Verbosity:** Simple statistical charts often require dozens of lines of “boilerplate” code to manage axes, colors, and legends manually
 - **Static nature:** It was primarily built for static, print-ready figures, making web-based interactivity difficult to implement

ALTAIR

- A **declarative** library: Abstract away the step-by-step logic,
 - allowing a user to declare the desired end state, while the library manages how to reach that state
- Designed for **rapid exploration** and concise code. For example,
 - Can generate complex, interactive charts with just a few lines by simply mapping data columns to visual channels (x, y, color)
- Built specifically to enable easy, **web-native interactivity** (tooltips, zooming, linking) that Matplotlib struggle with
- However, moving beyond the “standard” grammar,
 - not easy to change the library’s defaults to achieve a very specific, non-standard layout